What	Linear Algebra	Fourier Series
Vector Space	$\mathbb{R}^n$	Piecewise smooth 2π-periodic functions on $\mathbb{R}$
Inner Product	$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^{n} u_i v_i$	$\langle f(t), g(t) \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) g(t)\, dt$
Orthonormal Basis $\left(\mathbb{R}^3\right)$	$\{(1,0,0),(0,1,0),(0,0,1)\}$	$\{1, \cos mt, \sin nt\},\ n, m \in \mathbb{N} \backslash \{0\}$
Representation of a Vector in the Basis	$\mathbf{x} = \sum_{i=1}^{n} a_i \mathbf{e}_i$	$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt$
Coefficients are Projections	$a_i = \langle \mathbf{x}, \mathbf{e}_i \rangle$	$a_0 = \langle f(t), 1 \rangle$ $a_m = \langle f(t), \cos mt \rangle$ $b_m = \langle f(t), \sin mt \rangle$

Table 1: Vector Spaces: Linear Algebra vs. Fourier Series

Note: Some people think that the orthonormal basis for the Fourier series should be $\{\mathbf{1}, \sqrt{2}\cos mt, \sqrt{2}\sin nt\}$, since

$$\langle \mathbf{1}, \mathbf{1} \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} \mathbf{1}^2\, dt = \frac{1}{\pi} \int_{-\pi}^{\pi} dt = \frac{1}{\pi}\, t \Big|_{-\pi}^{\pi} = \frac{1}{\pi}\left(\pi - (-\pi)\right) = \frac{1}{\pi} 2\pi = 2$$

OTOH, $\langle \cos nt, \cos nt \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos^2 nt\, dt = \frac{1}{\pi}\pi = 1$. In the same way, $\langle \sin nt, \sin nt \rangle = 1$.

The one of the points here is that $\langle \hat{\mathbf{u}}, \hat{\mathbf{u}} \rangle = 1$ for all unit vectors $\hat{\mathbf{u}}$, so what about $\langle \mathbf{1}, \mathbf{1} \rangle$ if $\mathbf{1}$ is part of an orthonormal basis (again, since $\langle \mathbf{1}, \mathbf{1} \rangle = 2$)?

Note also that here $\langle \mathbf{1}, \sin nt \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} \mathbf{1} \sin nt\, dt = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin nt\, dt = 0$. Similarly, $\langle \mathbf{1}, \cos nt \rangle = \langle \cos nt, \sin mt \rangle = 0$.

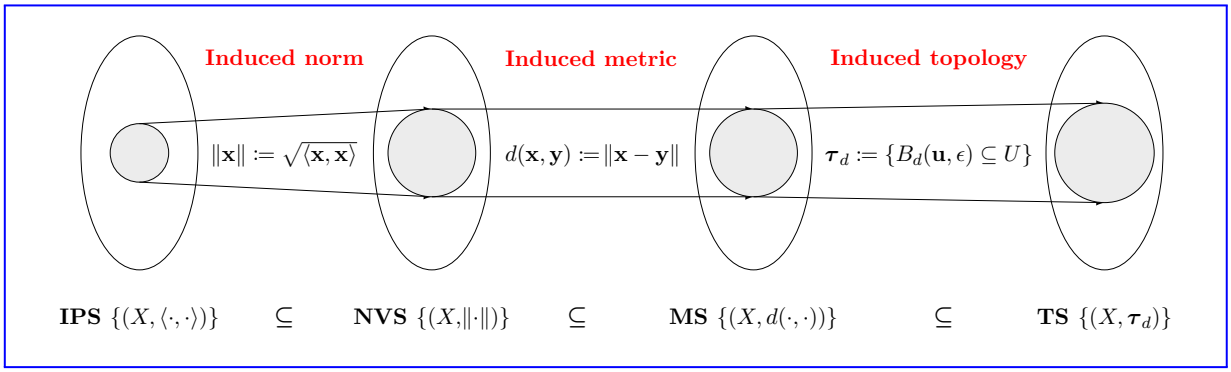

Figure 1: Induced Norm, Metric, and Topology

IPS	$--$	Inner Product Spaces	$(X, \langle \mathbf{x}, \mathbf{y} \rangle)$
NVS	$--$	Normed Vector Spaces	$(X, \|\mathbf{x}\|) = (X, \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle})$
MS	$--$	Metric Spaces	$(X, d(\mathbf{x}, \mathbf{y})) = (X, \|\mathbf{x} - \mathbf{y}\|) = (X, \sqrt{\langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y} \rangle})$
TS	$--$	Topological Spaces	$(X, \boldsymbol{\tau}_d)$, where $\boldsymbol{\tau}_d = \{\text{open subsets of } (X, d)\}$

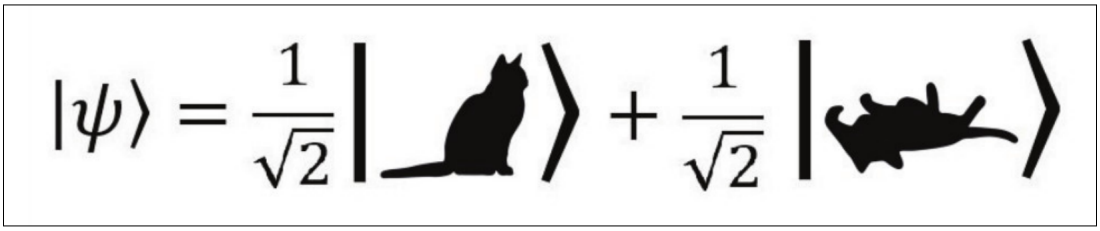

Figure 2: Quantum superposition and Schrödinger's cat

$n=\langle\ ,\ \rangle$
$G(\ )$
$\ \ \ \

0.6249999121401877

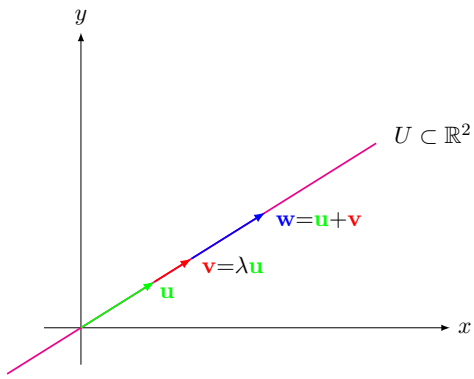

Figure 5: The line U is a linear subspace of $\mathbb{R}^2$

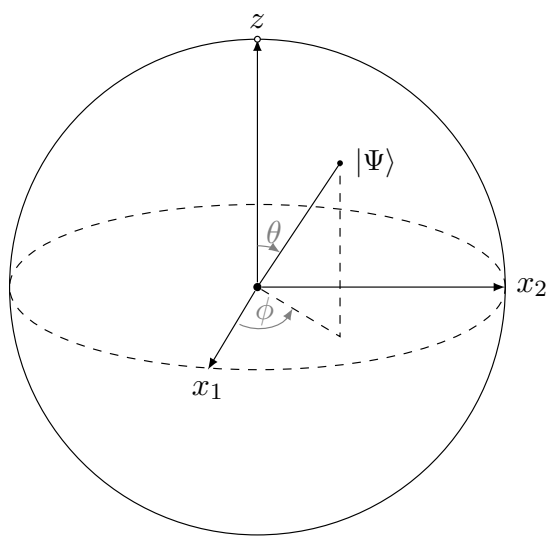

Figure 6: The Bloch Sphere

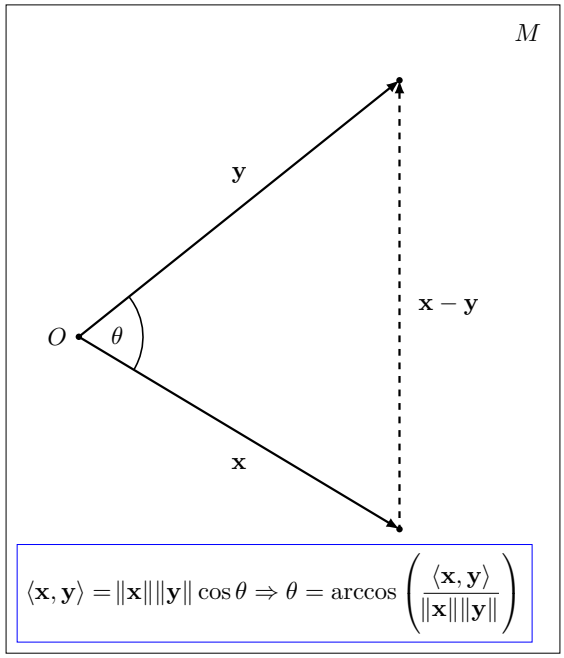

Figure 7: Geometric interpretation of $\langle \mathbf{x}, \mathbf{y} \rangle$

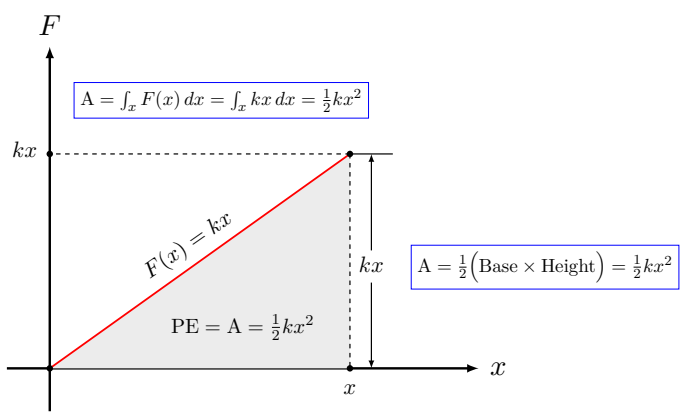

Figure 8: Potential Energy of a Simple Harmonic Oscillator

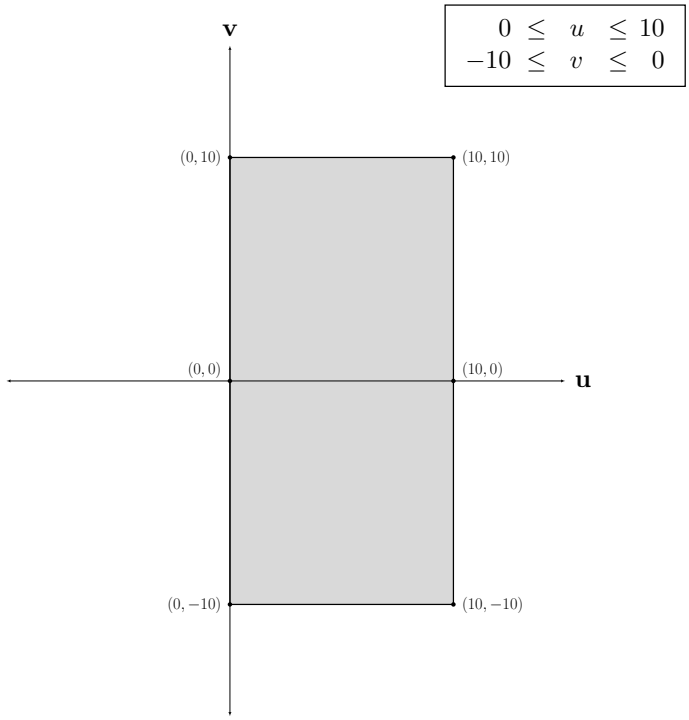

Figure 9: The transformed parallelogram is a rectangle in the uv-plane

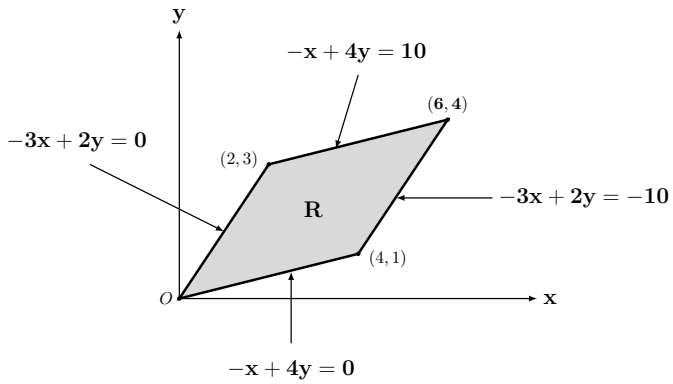

Figure 10: Equations of the Sides of the Parallelogram

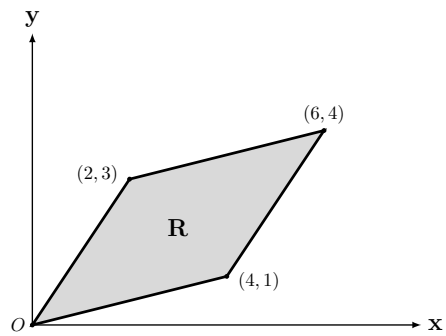

Figure 11: Parallelogram in the xy-plane

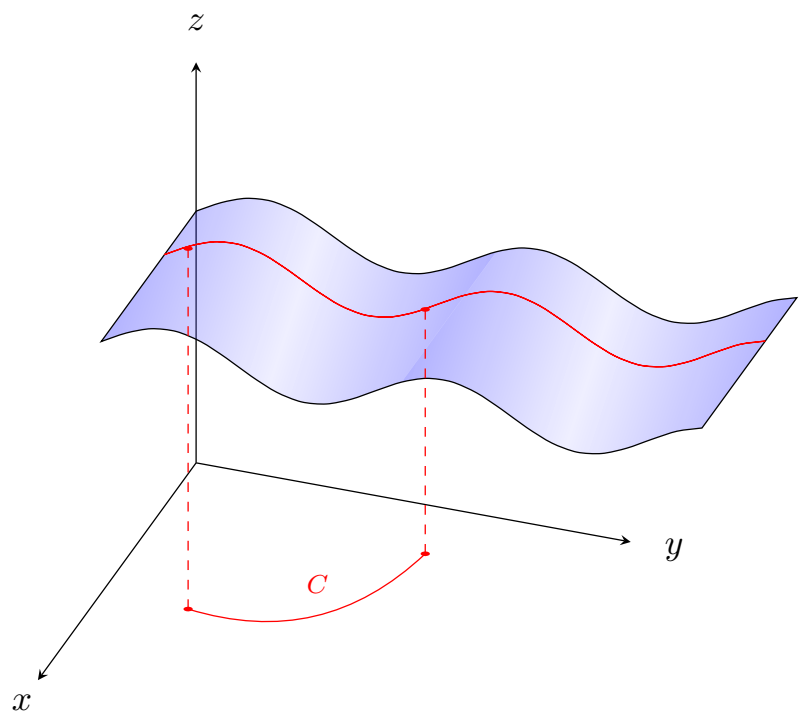

Figure 12: 3D

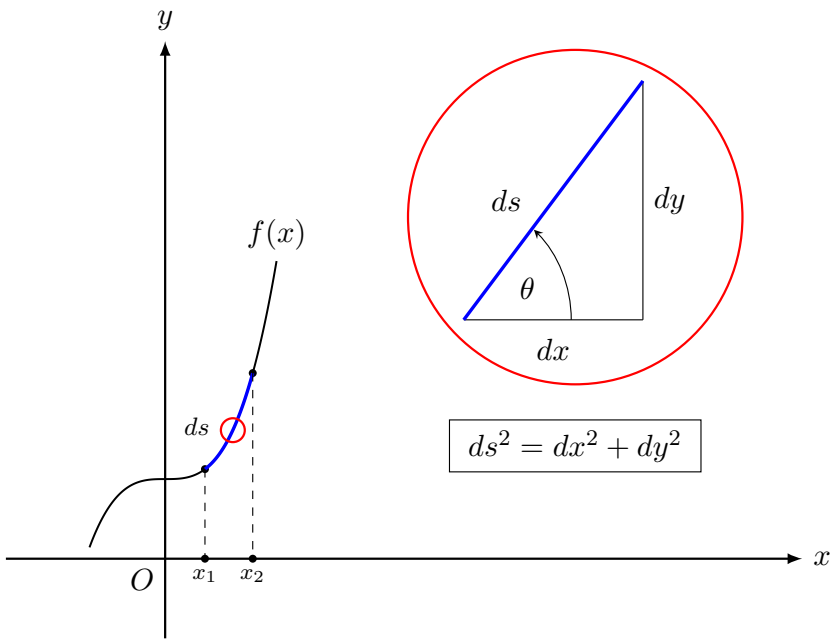

Figure 13: $f(x)$, ds and the Pythagorean Theorem

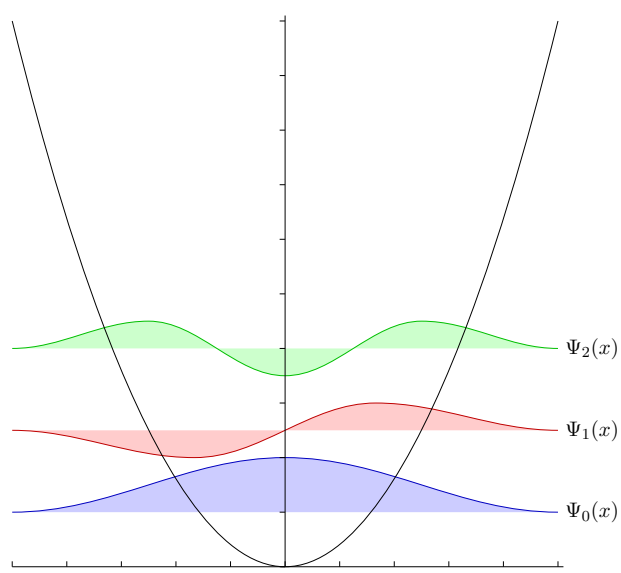

Figure 14: Wave Functions

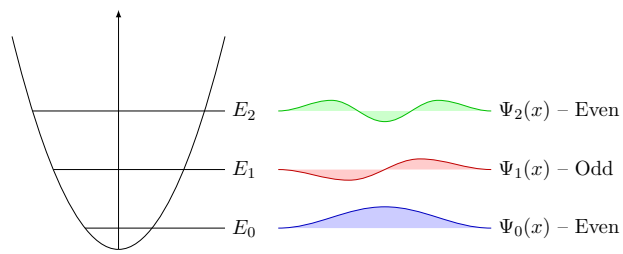

Figure 15: Wave Functions

9^2	$=$	81	$\longrightarrow$	$8+1$	$=$	9
45^2	$=$	2025	$\longrightarrow$	$20+25$	$=$	45
703^2	$=$	494209	$\longrightarrow$	$494+209$	$=$	703
7777^2	$=$	60481729	$\longrightarrow$	$6048+1729$	$=$	7777
857143^2	$=$	734694122449	$\longrightarrow$	$734694+122449$	$=$	857143

Figure 16: A Few Example Kaprekar Numbers

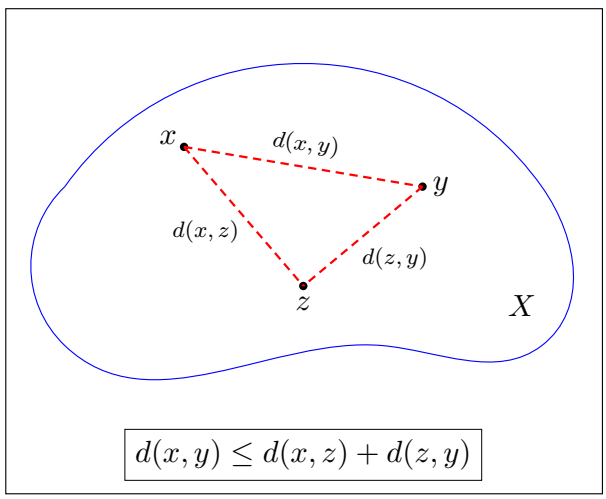

Figure 17: The Triangle Inequality

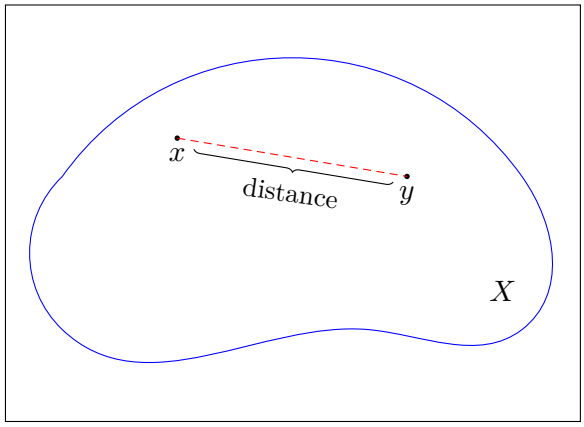

Figure 18: X, x and y

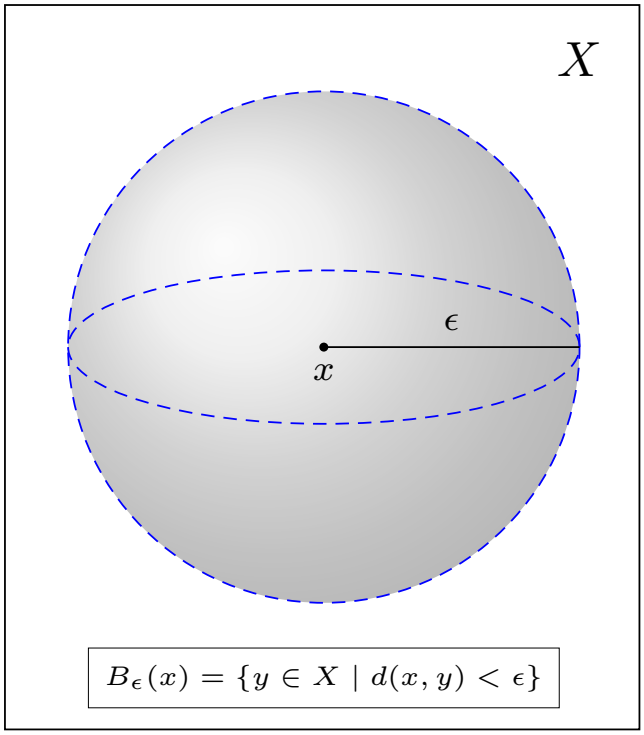

Figure 19: $B_\epsilon(x)$ is an open epsilon ball centered at x with radius ϵ

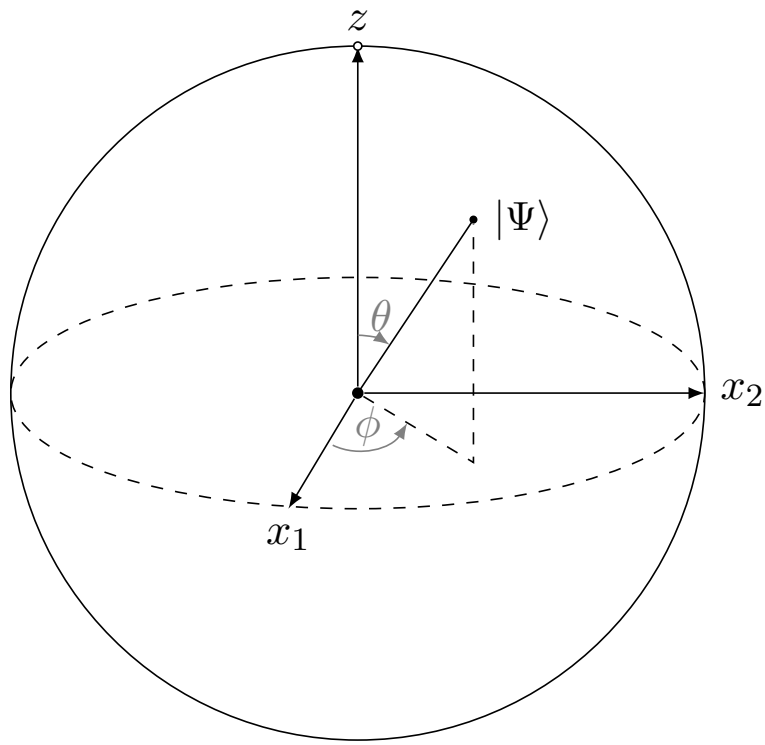

Figure 20: The Bloch Sphere

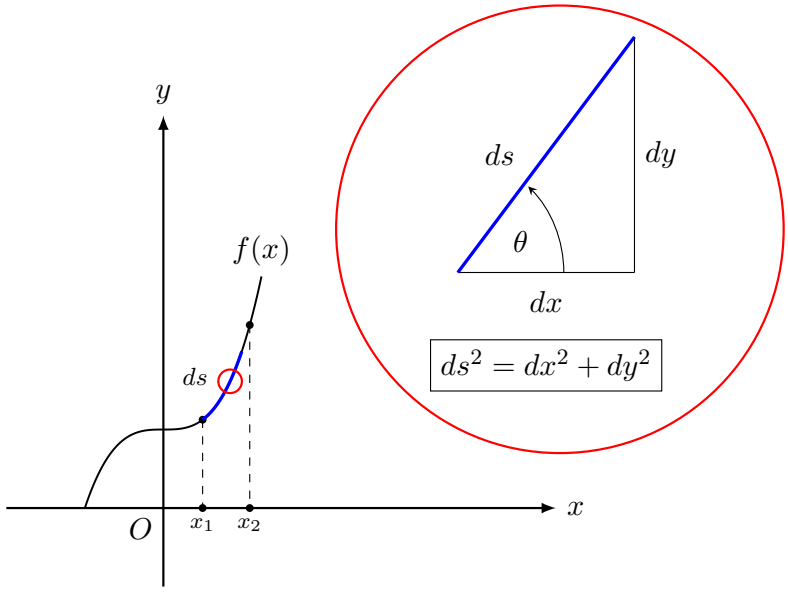

Figure 21: $f(x)$, ds and the Pythagorean Theorem

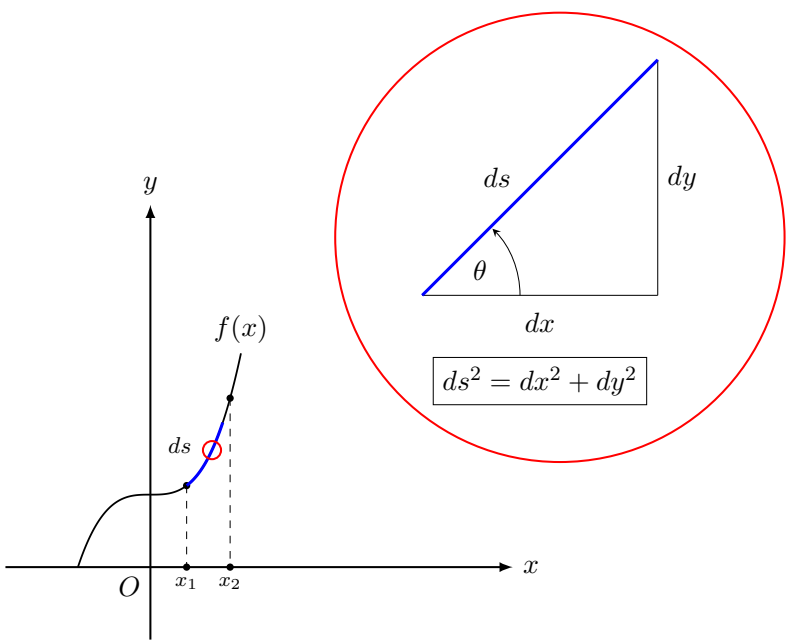

Figure 22: $f(x)$, ds and the Pythagorean Theorem